



**CANDIDATE
NAME**

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CLASS

2T

**INDEX
NUMBER**

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BIOLOGY

Paper 3

9747/03

Thursday 17 September 2009

1 hour 30 min

Additional materials: Answer Booklet / Paper

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

At the end of the examination,

1. fasten all your work securely together;
2. enter the number of the Section C question you have answered in the grid opposite.

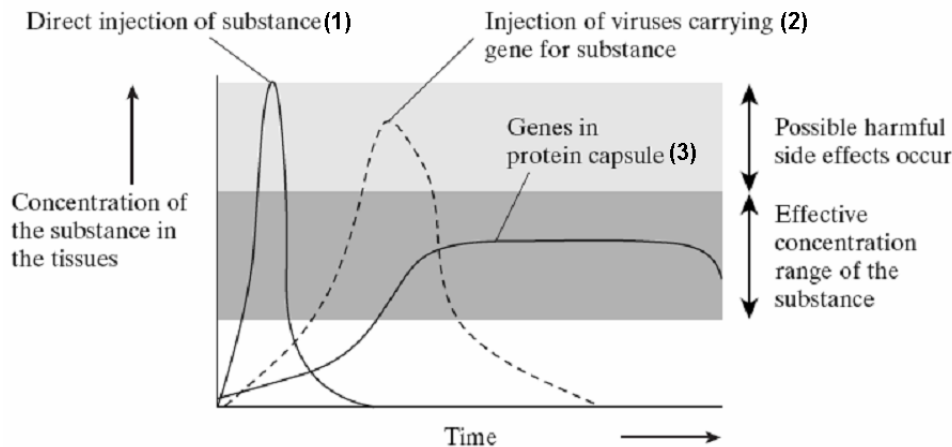
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [15]	
2 [15]	
3 [15]	
4 [20]	
Total	

- 1 In gene therapy, genes are introduced into a person who has defective genes which do not produce an important substance. Three experiments were done to compare techniques for introducing an important substance into a person with defective genes.

1. The substance was injected directly.
2. Harmless viruses carrying genes coding for the substance were injected.
3. The genes were put into a protein capsule which was inserted into the tissues.

The graph shows results of the experiments.



- (a) (i) State three key differences between graphs 2 and 3. [3]

1. **Gradient for the increase** in substance is steeper in graph 2 then in 3
2. **increase in substance** is faster in graph 2 then in 3
3. In 3, amount of time that the **concentration is in the effective range** is **4.5 times** that of 2

OR

1. In 3, amount of time that **the concentration is in the effective range** is **greater than in 2**.
2. **Concentration of substance reaches toxic concentrations** (harmful side effects occur) but in **2 but not in 3**. Concentration of substance remains below toxic levels in 3.
3. **Duration where substance remains** in body is **longer in 3 than in 2**

- (ii) Using the information in the graph, suggest **two** advantages and **one** disadvantage of the capsule method compared to the others. [3]

Advantages

1. substance never reaches harmful levels/no side effect/less likely to harm the organism;
2. longer relief from symptoms/ longer effective range/less frequent treatment needed

Disadvantage

3. takes longer to take effect

(b) (i) Some people are concerned about using gene therapy to treat genetic disorders in humans. Give **three** arguments against the treatment of disorders by gene therapy. [3]

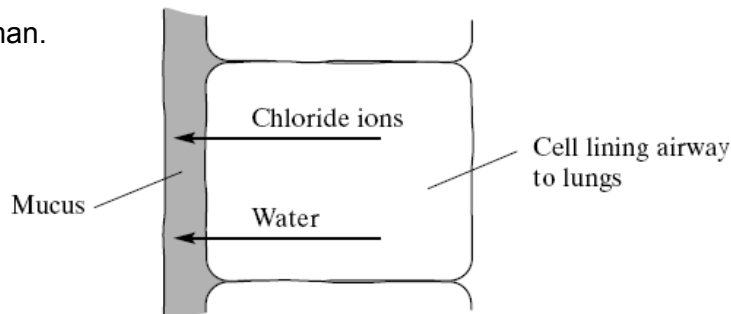
1. **high cost** plus reason e.g. compared with conventional treatments
2. risk of **stimulating the immune system** (by the vector)
3. use of **animals in (preliminary) testing**, other genes introduced which may have **damaging effects**/damage genome/damage other genes
4. may induce **a tumor if integrated in a tumor suppressor gene** because insertional mutagenesis → cancer may result.
5. **short-lived nature** of gene therapy so patients **will have to undergo multiple rounds** of gene therapy
6. **immune system's enhanced response** to invaders it has seen before makes it **difficult for gene therapy to be repeated in patients**
7. possibility of **viral vector recovering its ability to cause disease** once in host
8. **inadequate understanding** of biological interactions of vector and host
9. conditions or disorders that arise from mutations in a single gene are the best candidates for gene therapy, but some the most commonly occurring disorders are caused by the **combined effects of variations** in many genes. **Multigene disorders especially difficult to treat effectively using gene therapy**
10. changes to genetic make-up of individual/species/genome/future generations/germ line may affect normal development of offspring

Reject - not ethical (without explanation)

(ii) Explain why a person cured by gene therapy may still have children who suffer from the abnormality. [2]

1. gene therapy treats somatic cells and gametes/germ cells are not affected/do not take up the healthy gene
2. will still pass on the defective genes

(c) The diagram shows the movement of chloride ions and water in the cells of the lung of a healthy human.



(c) (i) In people with cystic fibrosis the gene controlling the movement of chloride ions is defective. The defective gene produces defective CFTR. Suggest why defective CFTR does not transfer chloride ions across cell membranes. [1]

1. **defective CFTR has different tertiary structure** / conformation / different shape so channel blocked / chloride ions do not fit

(ii) Identify the exact genetic nature of the disease CF [1]

1. Genes on **Chromosome 7** are affected
2. Mutation of this **recessive gene** is due to the deletion of 3 base pairs (codon # 508 in mRNA) from the CFTR gene, (**ΔF508**) → phenylalanine to be missing (though in 15% of the cases, cystic fibrosis is caused by five other mutations).

(iii) Describe briefly **one** way in which normal CFTR genes might be introduced into the cells lining the airways of the lungs. [2]

1. Liposomes: Functional CFTR gene into liposomes directly / into plasmids into liposomes;
2. Use of aerosol spray / inhalers to deliver vector
3. liposomes fuse with membrane of cells delivering genes across membrane into cells

OR

1. Virus: Functional CFTR gene into attenuated virus
2. Use of aerosol/sprays/inhalers to deliver vector
3. Virus infects cells delivering genes across membrane into cells

[Total : 15]

2 A Restriction Fragment Length Polymorphism (RFLP) locus is very closely linked to an autosomal disease-causing gene.

When an individual's DNA sample is digested with *EcoRI*, it produces two bands of interest, a 7kb and an 8kb band on the Southern blot.

(a) Explain how Restriction Fragment Length Polymorphisms come about. [2]

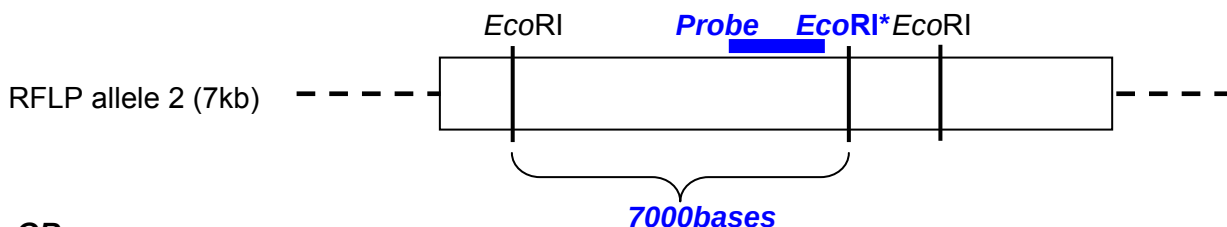
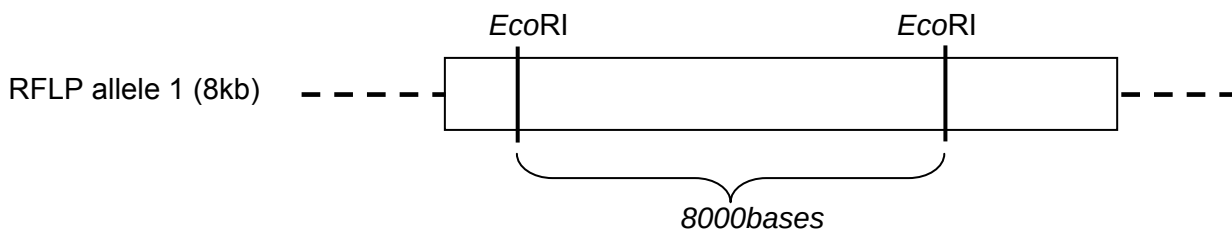
1. **Difference in nucleotide sequence** between chromosomes / individuals
2. Causes **palindromic sites** for a particular **RE to occur at different places** along genome
3. When cut with a **particular (Same) Restriction Enzyme**
4. Will produce **Restriction fragments of differing lengths**

(b) If the RFLP allele causing the 8kb band has the *EcoRI* restriction sites as shown below, **draw on the RFLP allele that is responsible for the 7kb band**, indicating:

- i) the position of the polymorphic restriction site on the 7kb allele, marking the polymorphic restriction site with an asterisk (*) and
- ii) the position of the probe.

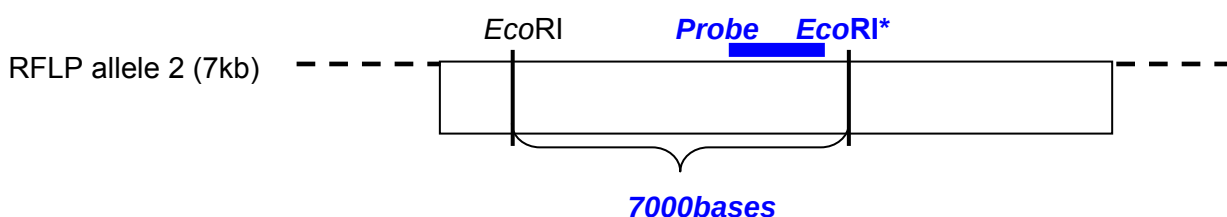
Label any important regions showing the distance between restriction sites as is demonstrated for you in the RFLP 8kb allele.

Dashes show the continuation of the rest of the chromosome. [2]



[2]

OR

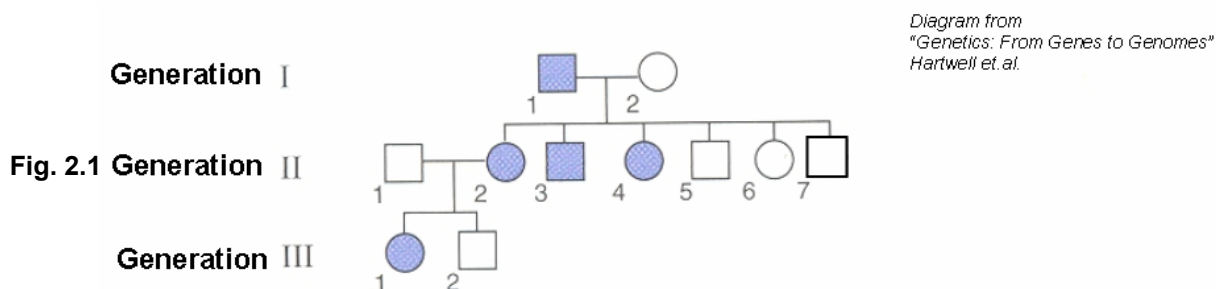


Probe can be anywhere along the region of different fragment length. Probes not completely overlapping the region are also accepted.

Candidates can choose to shift one of the two EcoRI sites on allele 1 to be the 'polymorphic restriction site', although by definition polymorphic restriction site simply means 'present on one allele absent on the other' thus it would be better to have a third EcoRI site. It is possible because the question says – "portion of Southern blot" and therefore it is possible that the band for 1kb does exist but is not shown.

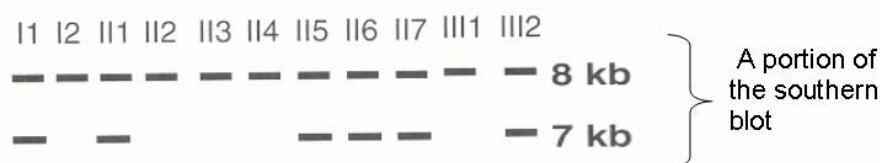
Figure 2.1 below shows the pedigree of a family in which the same autosomal disease-causing dominant allele is transmitted through two generations. The figure also shows a corresponding portion of a Southern blot with DNA samples from each individual in the family, digested with EcoRI and probed with a DNA fragment that detects a RFLP.

Squares indicate males and circles indicate females, shaded squares and/or circles indicate that the individual is affected by the disease.



Restriction fragment digests of DNA from individuals.

Roman numerals I / II / III indicate the generation that the individual is from and 1-7 indicate the individual numbered in the pedigree chart.



With reference to the Fig. 2.1, examine the pattern of inheritance of the disease-causing dominant allele and the RFLP 8kb and 7kb alleles in the pedigree chart.

(c) (i) Which allele, allele 1(8kb) or allele 2 (7kb) is always inherited together with the disease-causing allele? [1]

1. 8kb allele

(ii) With reference to the transmission of alleles from Generation I to Generation II, explain your answer in (c)(i). [3]

1. **II2, II3 and II4 are affected** and are **homozygous for the 8kb allele**.
2. Thus they must have received **one 8kb allele from their father I1** who is affected and thus also has the disease causing allele
3. **II5, II6 and II7 must have received the 7kb allele from father I1** because the **mother is homozygous for the 8kb allele**. They **are not affected**, and thus **do not have the disease causing allele**.

Answer that uses pattern of "most of those with 8kb allele also have disease" will not be accepted because I2, II5-7 also have 8kb allele but do not suffer from the disease.

(d) Outline the steps required to get from the binding of probe to the image shown in the southern blot. [2]

1. **Probe that is on the rest of the nitrocellulose membrane is then washed off**, leaving the probe bound only to the complementary DNA
2. Depending on what kind of probe was used, **radioactive probe – autoradiography** will be used to produce the image shown, or for **fluorescent probes – UV light** will be shone onto the membrane to produce the image shown.

(e) Explain how such RFLP markers have aided in genomic mapping. [3]

1. **Increased number of markers** besides gene markers
2. **Non-coding regions** which may **make up a large proportion of the genome, can be mapped**
3. **Distances in terms of nucleotides can be determined from restriction fragment lengths**

(f) How is genomic mapping advantageous as compared to obtaining the sequence of all the genes in an organism? [2]

1. Relative position of genes may not be known from the sequence alone, thus **genomic mapping allows for both position and sequence to be known**
2. Regulatory sequences can be **studied and therefore mechanisms for the switching on / off of the genes can be better known.**

[Total 15]

(NJC Prelim 08/P3/3)

- 3 Equal masses of tobacco plant callus were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin.

The results are shown in Table 3.1.

Table 3.1

treatment	concentration of plant growth regulators / mg dm ⁻³		effect of plant growth regulators on callus growth
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

(a) Explain why growing plantlets from a callus in tissue culture results in a clone. [3]

1. *This is a type of asexual reproduction.*
2. *All the plantlets are genetically identical / have the same genotype because they are produced by mitotic division.*
3. *The process of mitosis involves exact replication of the parental DNA.*

(b) State why the plant tissue cultures are initially grown in sterile conditions. [1]

1. To prevent **contamination by bacteria or fungi** (which can grow very rapidly on the nutrient media).

(c) With reference to Table 3.1,

(c) (i) Describe the effects of auxin and cytokinin on callus growth. [3]

1. Both hormones need to be present for any growth to occur.
2. The type of growth depends on the relative concentrations of the auxin and cytokinin.
3. A low level of cytokinin (0.02 mg dm⁻³) to auxin (2.00 mg dm⁻³) triggers root growth.
4. Increasing the relative level of cytokinin (0.50 mg dm⁻³) to auxin (2.00 mg dm⁻³) triggers shoot growth.

(ii) Suggest what treatment should be given to the shoots developed from callus in treatment D to turn them into plantlets. [2]

1. From the table, transferring the shoots from treatment D to treatment B /
2. changing the cytokinin levels from 0.50 mg dm⁻³ to 0.02 mg dm⁻³ would stimulate the growth of roots and the developments of plantlets.

(d) Outline, using **one** named example, the process of genetic engineering that could increase crop yield. [4]

1. Any named example, insect-resistant (Bt) / herbicide-tolerant crop (cotton, corn, soybean, tomato plant)

Process of genetic engineering:

2. The gene for Bt toxin (from *Bacillus thuringiensis*) / the gene encoding glyphosate-resistance is isolated.
3. The gene of interest is joined to a tumour-inducing / Ti plasmid that has been cut with the same restriction enzyme using DNA ligase.
4. The resultant recombinant plasmid is transformed into *Agrobacterium tumefaciens*.
5. Selection for *Agrobacterium* cells that have successfully taken up the recombinant plasmid, which are then cultured with leaf discs / callus tissue to mediate the transfer of the recombinant plasmid.
6. Alternatively, the gene of interest can be inserted into plant tissue using a gene gun.

Explanation how growing the genetically modified crop can lead to an increase in crop yield:

7. Bt crops are protected against damage by insect pests, so farmers have higher crop yield.
8. Farmers may apply a specific herbicide to control weeds population, without damaging their herbicide-tolerant crops. There will be a higher yield because of less competition from weeds.

(e) Identify **one** possible hazard and benefit of plant genetic engineering. [2]

Possible hazard(s):

1. Emergence of Bt toxin-resistant insects (by natural selection)
2. Genetically modified crop becomes a "weed" that out-competes wild crops for nutrients, space etc.
3. Possible transfer (via pollination) of the herbicide-resistant gene into weeds, creating weeds that are resistant to herbicides.
4. Accidental incorporation of a antibiotic selection marker into a pathogenic bacteria might result in antibiotic resistance of that bacterial species.

Possible benefit(s):

1. Economic benefit to farmers (either higher crop yield, lower output as no need to purchase pesticides / herbicides etc.)
2. Health benefit to consumers (because farmers use less pesticides).

[Total: 15]

- 4 (a)** Compare the mutations responsible for cystic fibrosis (CF), X-linked and ADA-deficient severe combined immunodeficiency (SCID) and comment on their pattern of inheritance.[10]

	X-linked SCID	ADA-deficiency SCID	Cystic fibrosis (CF)	Marks
Inheritance	X-linked	Autosomal recessive	Autosomal recessive	1
Chromosome affected	X chromosome	Chromosome 20 of ADA gene	Chromosome 7 of CFTR gene	1
Clinical symptoms	Absence of infection fighting cells i.e. T and B cells. Patients will be prone to virus and bacterial infection	Absence of infection fighting cells i.e. T and B cells. Patients will be prone to virus and bacterial infection	Accumulation of sticky mucus, leading to chronic infection	

Pattern of inheritance for X-linked SCID:

1. Thus, a female who is a carrier for X-linked SCID may show no signs of immunodeficiency
2. Her sons and daughters have a 50 percent chance of inheriting the X chromosome with the mutation
3. The son who has inherited one copy of the mutant X chromosome will have X-linked SCID
4. The daughter who has inherited one copy of the mutant X chromosome will be a carrier like their mother

Pattern of inheritance for ADA-deficient SCID:

5. Sufferers must be homozygotes since it is a recessive disorder
6. Meaning both parents (if carriers) must pass on the defective gene for any of their children to get the disease
7. Heterozygotes are carriers as they carry only one copy of the faulty gene
8. The chances of 2 carriers having a child with cystic fibrosis are 1 in 4

Pattern of inheritance for cystic fibrosis:

9. Sufferers must be homozygotes since it is a recessive disorder
10. Meaning both parents (if carriers) must pass on the defective gene for any of their children to get the disease
11. Heterozygotes are carriers as they carry only one copy of the faulty gene
12. The chances of 2 carriers having a child with cystic fibrosis are 1 in 4

- (b) (i)** Discuss the benefits and ethical concerns of the Human Genome Project [10]

1. provides a detailed physical, genetic map of the human genome and DNA sequence of entire human genome
2. allow elucidation of genes associated with human diseases/ understanding genetics basis of disease;
3. allows early detection or diagnosis via genetic testing;
4. e.g. risk of breast cancer, diabetes and colon cancer;
5. improve diagnosis of diseases prenatally;
6. pharmacogenomics : design customized drugs based on genetic profile;
7. enables novel treatment through gene therapy as gene sequences are now readily available;
8. pharmaceutical products such as medicine can be synthesized using readily available gene sequences; e.g. humulin and tissue plasminogen activator (TPA) given to after heart attack to dissolve blood clots;
9. a means of risk assessment to evaluate health risks of individuals who may be exposed to radiation or carcinogens;

max : 6 points

10. ethical issues of pre-implantation genetic testing viewed as “tampering with nature” by certain religious groups;
11. To-be parents may have to make difficult decisions of terminating pregnancy due to presence of genetic disorder. It may amount to murder since there are different definitions of when a foetus is recognized as living;
12. positive test result indicates that a person possesses a known mutation and may not necessarily develop cancer. Are the general population informed adequately;
13. management of genetic information, who owns the right to the genetic information? The person sample taken from or the hospital or company who sequenced the information?
14. work-related discrimination: is it right for employers to demand DNA testing because it can reveal any possible high risk disease the employee may develop and thus increase costs of employing this person;
15. insurance-related discrimination: is it right for insurance companies to demand DNA testing because it can reveal any possible high risk disease which is not covered under the insurance policy;
16. Designer drugs developed by pharmacogenomics discriminates against the poor as only the wealthy have the ability to pay for such services;

Max: 4 points

Table of Specs

	Structured			Essay	
	Recall	Thinking	Higher Thinking	Thinking	Higher Thinking
Isolating Cloning and Sequencing DNA					
H1] DNA Cloning (GE)	1a, 2a, 2b	1b, 2c			
H2] Analysis n Genomics, RFLP/ Gel electro / PCR					
H3] Human Genome Project					
Applications of Molecular and Cell Biology					
I1] Stem Cells					
I2] Treatment of SCID / CF		1c		4a	4b
I3] Gene Therapy					
I4] Cloning in Plants	3a, 3b	3c, 3d,	3e		
I5] GE organisms					
Total					